

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 13A.3

FACTORS AFFECTING EQUILIBRIUM CONSTANTS 1

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	13: CHEMICAL EQUILIBRIA
SUB-TOPIC:	13A.3: Factors Affecting Equilibrium Constants 1
TOTAL QUESTIONS:	3 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	8 Marks
EXAMINER FAQs:	2 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 13A.3

Q1. 1. The synthesis of ammonia is an exothermic reaction:

$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$ Explain, in terms of K_p , what happens to the equilibrium yield of ammonia when the temperature is increased.

[4 Marks]

WCH14/01/Jan23/Q5

Q2. 2. For an endothermic reaction, how does an increase in temperature affect the value of K_c ?

[1 Marks]

WCH14/01/Oct22/Q7

- A. K_c increases
- B. K_c decreases
- C. K_c remains constant
- D. K_c becomes zero

Q3. 3. An equilibrium mixture of NO_2 (brown) and N_2O_4 (colourless) is heated. The mixture becomes darker brown. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ Deduce the sign of ΔH for the forward reaction, explaining your reasoning based on K_p .

[3 Marks]

WCH14/01/Jun21/Q6

SECTION B: WORKED MARK SCHEME — 13A.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. The synthesis of ammonia is an exothermic reaction: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ $\Delta H = -92 \text{ kJ mol}^{-1}$ Explain, in terms of K_p, what happens to the equilibrium yield of ammonia when the temperature is increased. 1. Forward reaction is exothermic, so K_p decreases as temperature increases. 2. The ratio $\frac{p(\text{NH}_3)^2}{p(\text{N}_2) \cdot p(\text{H}_2)^3}$ must decrease to equal the new K_p. 3. The position of equilibrium shifts to the left. 4. Therefore, the equilibrium yield of ammonia decreases.	[4]
Q2	Q2: 2. For an endothermic reaction, how does an increase in temperature affect the value of K_c? A	[1]
Q3	Q3: 3. An equilibrium mixture of NO_2 (brown) and N_2O_4 (colourless) is heated. The mixture becomes darker brown. $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ Deduce the sign of ΔH for the forward reaction, explaining your reasoning based on K_p. 1. Mixture becomes darker, so equilibrium shifted to the right (more NO_2). 2. This means K_p has increased with temperature. 3. Therefore, the forward reaction must be endothermic (ΔH is positive).	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 13A.3**FAQ #1: Why must I explain the effect using K_p/K_c rather than Le Chatelier's principle?****Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Why must I explain the effect using K_p/K_c rather than Le Chatelier's principle?

MODEL ANSWER / EXAMINER EXPECTATION:

If the question explicitly asks you to explain "in terms of K_c/K_p ", using Le Chatelier's principle alone will score zero. You must state how the constant changes and how the concentration/pressure quotient adjusts to meet the new constant.

FAQ #2: Does temperature always change the equilibrium constant?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Does temperature always change the equilibrium constant?

MODEL ANSWER / EXAMINER EXPECTATION:

Yes, temperature is the ONLY factor that changes the value of the equilibrium constant (K_c or K_p). Pressure, concentration, and catalysts do not.